

Quantum SVM Crash Report

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Summary:

While running the Quantum SVM implementation using Qiskit Aer backend, the system encountered an 'unknown software exception' which caused Visual Studio Code to shut down unexpectedly. The error is suspected to originate from the native C++/CUDA layer in Qiskit Aer when using the statevector simulator backend.

Possible Causes:

Incompatible GPU driver or CUDA version with Qiskit Aer.
Mismatched Qiskit, qiskit-aer, and qiskit-machine-learning versions.
Deprecated QuantumKernel usage instead of FidelityQuantumKernel.
Excessive dataset size causing memory overload in quantum simulator.

Steps Taken:

Replaced deprecated QuantumKernel with FidelityQuantumKernel.
Introduced ComputeUncompute fidelity calculation for better stability.
Limited dataset to n_samples=200 to prevent simulator memory issues.
Removed duplicate imports and deprecated API calls.
Suggested updating Qiskit and Qiskit Aer to latest versions.

Recommended Actions:

Update Qiskit packages using: <code>pip install --upgrade qiskit qiskit-aer qiskit-machine-learning</code>
Test using a pure-Python quantum simulator backend to avoid native crashes.
Consider running the quantum simulation inside Docker or conda for isolation.
If GPU is enabled, verify CUDA and driver compatibility.